

MODULE 1

ENVIRONMENT

“**Environment**” comes from the old French words **environ** and **en-vironer**, which mean “**to surround**”, “**to enclose**”, or “**to encircle**”. Environment is the surroundings and conditions in which living organisms, like plants, animals, and humans, live. It is made up of both **biotic (living)** and **abiotic (non-living)** things.

1. Abiotic things

- The abiotic environmental components include basic:
- **Inorganic elements and compounds** such as **water and carbon dioxide, calcium and oxygen, carbonates and phosphates**
- **Physical factors** as **soil, rainfall, temperature, moisture, winds, currents, and solar radiation** with its concomitants of light and heat

2. Biotic things

- The biotic environmental factors comprise **plants, animals, and microbes**.
- They interact in a fundamentally energy-dependent fashion

Environment includes water, air and land and the interrelation ships which exist among and between water, air and land and human beings and other living creatures such as plants, animals and micro-organisms.

Components of Environment

Environment has been classified into four major components:

1. Hydrosphere
2. Lithosphere
3. Atmosphere
4. Biosphere

Hydrosphere includes all water bodies such as lakes, ponds, rivers, streams and ocean etc. Hydrosphere functions in a cyclic nature, which is termed as hydrological cycle or water cycle. Lithosphere means the mantle of rocks constituting the earth’s crust. The earth is a cold spherical solid planet of the solar system, which spins in its axis and revolves around the sun at a certain constant distance.

Lithosphere mainly, contains soil, earth rocks, mountain etc. Lithosphere is divided into three layers-crusts, mantle and core (outer and inner). Atmosphere The cover of the air, that envelope the earth is known as the atmosphere.

Atmosphere is a thin layer which contains gases like oxygen, carbon dioxide etc. and which protects the solid earth and human beings from the harmful radiations of the sun.

There are five concentric layers within the atmosphere, which can be differentiated on the basis of temperature and each layer has its own characteristics. These include the troposphere, the stratosphere, the mesosphere, the thermosphere and the exosphere.

Biosphere it is otherwise known as the life layer, it refers to all organisms on the earth's surface and their interaction with water and air. It consists of plants, animals and micro-organisms, ranging from the tiniest microscopic organism to the largest whales in the sea.

INTRODUCTION - ENVIRONMENTAL SCIENCE

- Environmental science is the study of how living things, like plants, animals, and humans, interact with the environment, including air, water, soil, and climate
- It focuses on understanding and solving problems like pollution, deforestation, and climate change to protect the Earth and make it a better place for all living things
- Environmental scientists work on subjects like the understanding of earth processes, evaluating alternative energy systems, pollution control and mitigation, natural resource management, and the effects of global climate change. Environmental issues almost always include an interaction of physical, chemical, and biological processes.

SCOPE OF ENVIRONMENTAL SCIENCE

Environment is not limited to issues of sanitation and health but it is now concerned with pollution control, biodiversity conservation, waste management and conservation of natural resources. This requires expert eyes and hence is creating new job opportunities. The opportunities in this field are immense not only for scientists but also for engineers, biologists. There is a good chance of opportunity to find a job in this field as environmental journalists.

Environmental science can be applied in the following spheres:

1.Ecosystem Structure and Function

The study of ecosystems mainly consists of the study of the processes that link the living organism or in other words biotic component to the non-living organism or a biotic component. So, for the study of environment, we should aware with biotic and a biotic component.

2. Natural Resource Conservation

For managing and maintenance of forests which are natural resources and for the maintenance of wildlife forms task under natural resource conservation. It is also a scope of environmental studies

3.Environmental Pollution Control

With the knowledge of environmental science everybody can control the pollution. People

can handle the waste management and also look for ways to control pollution on the aspect of pollution control.

4. Environmental management

There are several independent environmental consultants who are working with Central and State pollution control Board. They offer advice to solve the problems of environment the optimum solution for the upcoming problems. They give direction for controlling pollution due to industrial development. There are several current consultants who are working with government pollution control boards, involved in policy making, pollution control and maintenance of ecological balance.

5. Scope of environmental studies in industry

Environmental scientist's work towards maintenance of ecological balance, they also work towards conservation of biodiversity and regulation of natural resources as well as on preservation of natural resources. Most of the industries have a separate environmental research and development section. These sections govern the impact that their industry has on the environment. Our environment is being degraded by the rapid industrialization.

6. Research and development

Research and development have tremendous scope due to increment in public awareness regarding the environment. Various universities and governmental organizations offer a scope for such research. These universities conduct research studies in order to develop the methods toward monitoring and controlling the source of environmental pollution. Due to an increasing threat from global warming, many steps are being undertaken for the reduction of greenhouse gases and the adoption of renewable energy resources. They generate awareness now regarding the use of solar energy for variety of purposes. This provides scope of environmental history in the field of research and development.

7. Social Development

NGO (Nongovernmental organizations) help in creating awareness regarding the protection of the environment and making the masses aware of various environmental issues. They also generate a public opinion in this field. They work towards disseminating information and in bringing about changes in political policies that are personally affect the environment. The social dimension of this profession includes controlling population explosion through organizing advisory awareness camps.

INTERDISCIPLINARY NATURE OF THE FIELD

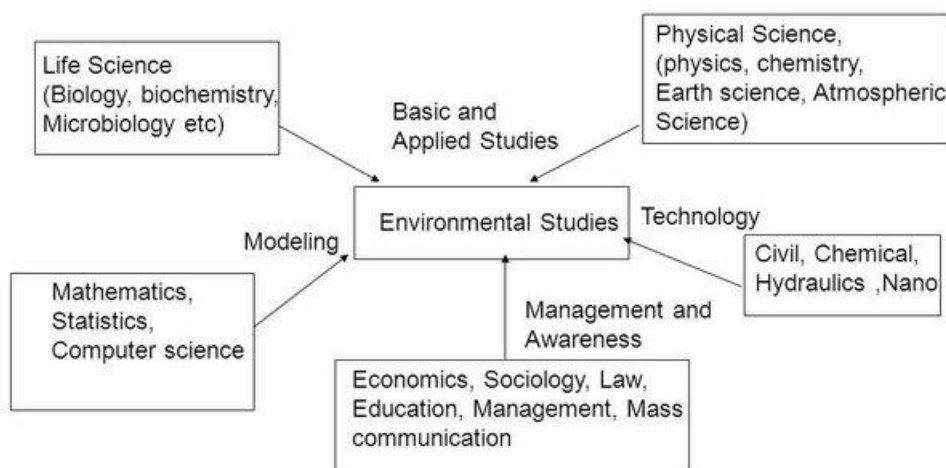
The **interdisciplinary nature of Environmental Science** means it combines knowledge from many different fields to solve environmental problems. For example:

- **Biology** - helps us understand plants, animals, and ecosystems
- **Chemistry** - helps us study pollutants and chemical processes in nature
- **Physics** - explains energy, climate, and natural forces
- **Geology** - studies the Earth's structure and natural resources
- **Social Sciences** - focus on how humans interact with the environment

Environmental science is an interdisciplinary academic field that integrates physical and biological sciences, (including but not limited to Ecology, Physics, Chemistry, Biology, Soil Science, Geology, Atmospheric Science and Geography) to the study of the environment, and the solution of environmental problems. Environmental science provides an integrated, quantitative, and interdisciplinary approach to the study of environmental systems.

Related areas of study include environmental studies and environmental engineering. Environmental studies incorporate more of the social sciences for understanding human relationships, perceptions and policies towards the environment. Environmental engineering focuses on design and technology for improving environmental quality.

The multidisciplinary nature of environmental science is illustrated in following diagram



ECOLOGY

- “The **scientific study** of the interactions of organisms with their physical environment and with each other, is called ecology”.

OR

- **Ecology** is the scientific **study of ecosystems** and the relationships between living organisms and their environment.
- It includes the study of how organisms interact with each other and their surroundings.

ECOSYSTEM

The terms ecosystems are most preferred, where ‘**eco**’ implies the **environment**, and ‘**system**’ implies an **interacting, inter-dependent** complex. The eco-system can be defined as any **spatial or organizational unit** including living organisms and non-living substances interacting **to produce an exchange of materials between the living and non-living parts**. The eco-system can be studied from either structural or functional aspects.

1. **Structural Aspect** - The structural aspects of ecosystem include a description of the **arrangement, types and numbers of species** and their life histories, along with a description of the physical features of the environment.
2. **Functional Aspect** - The functional aspects of the ecosystem include the **flow of energy and the cycling of nutrients**

Structural point of view all ecosystems consist of following two basic components:

1. **Abiotic Substances (Non-Living Components)** - The Abiotic substances include basic inorganic and organic compounds of the environment or habitat of the organism.
 - (a) **Inorganic Components:** The inorganic components of an ecosystem are as under carbon dioxide, water, nitrogen, calcium, and phosphate. All of these are involved in matter cycles (biogeochemical cycles)
 - (b) **Organic Components:** The organic components of an ecosystem are proteins, carbohydrates, lipids and amino acids, all of these are synthesized by the biota (flora and fauna) of an ecosystem and are reached to ecosystem as their wastes, dead remains, etc.
 - (c) The climate, temperature, light, soil etc., are other abiotic components of the eco-system.
2. **Biotic Substances (Living Components)** - This is indeed the trophic structure of any ecosystem, where living organisms are distinguished on the basis of their nutritional relationships.

Different types of ecosystems – (its structural and functional aspects with examples)

1. Forest ecosystem
2. Grassland ecosystems
3. Desert ecosystem
4. Aquatic ecosystems (ponds, lakes, streams, rivers, estuaries, oceans)

Division of Ecosystem

The ecosystem can be divided, from the energetic view point into three types of organisms:

1. **Producer** - Photosynthetic algae, plants and bacteria are the producers of the ecosystem; all other organisms depend upon them directly or indirectly for food
2. **Consumers** - Consumers are herbivorous, carnivorous, and omnivorous animals; they eat the organic matter produced by other organisms
3. **Reducers** - Reducers are heterotrophic organisms like animals; they are fungi and bacterial that decompose dead organic matter

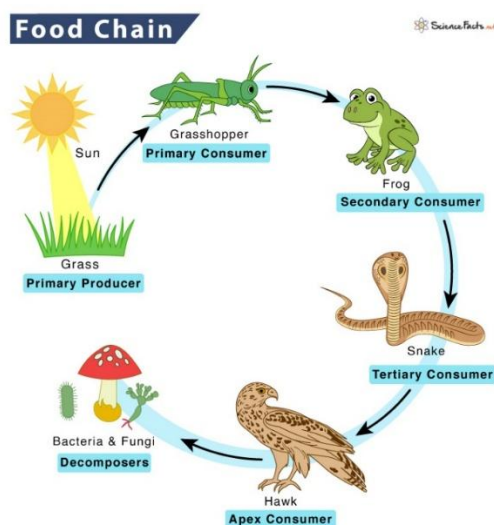
Functions of Eco-system

The functions of Ecosystem are as under:

1. Transformation of Solar Energy into Food Energy
2. The Circulation of elements through Energy Flow
3. The Conversion of Elements into Inorganic Flow
4. The Growth and Development of Plant
5. Productivity of ecosystem

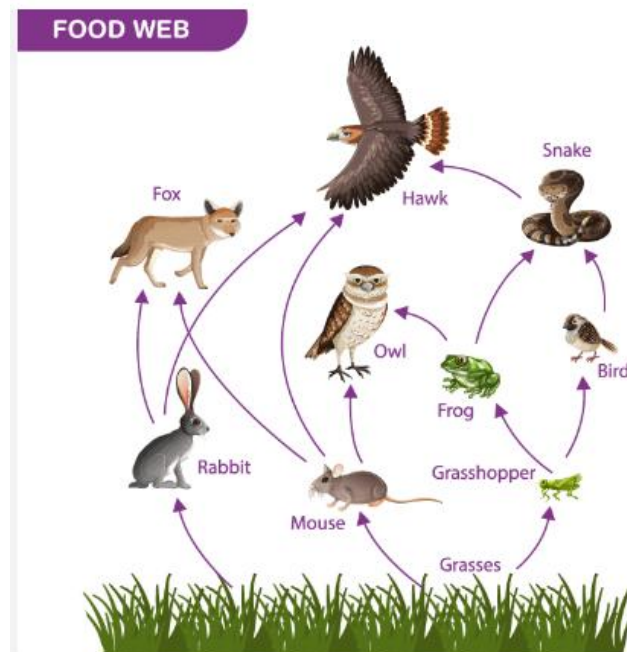
FOOD CHAINS

The most obvious aspect of nature is that **energy must pass from one living organism to another**. When herbivorous animals feed on plants, energy is transferred from plants to animals. The transfer of energy from the source in plants through a series of organisms by eating and being eaten constitutes **food chains**. Each chain usually has only **four to five** such links. However, a single species may be linked to a large number of species.



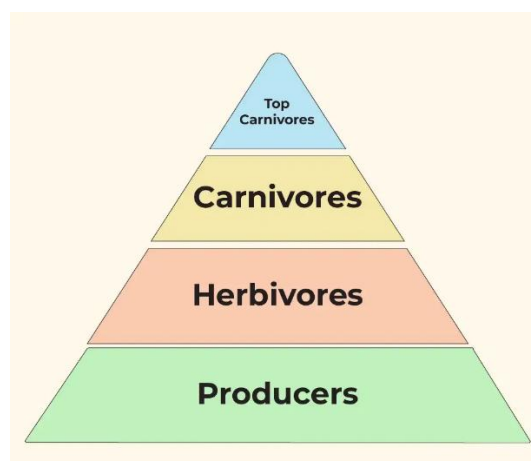
FOOD WEBS

In an ecosystem there are a very large number of interlinked chains. This forms a food web.



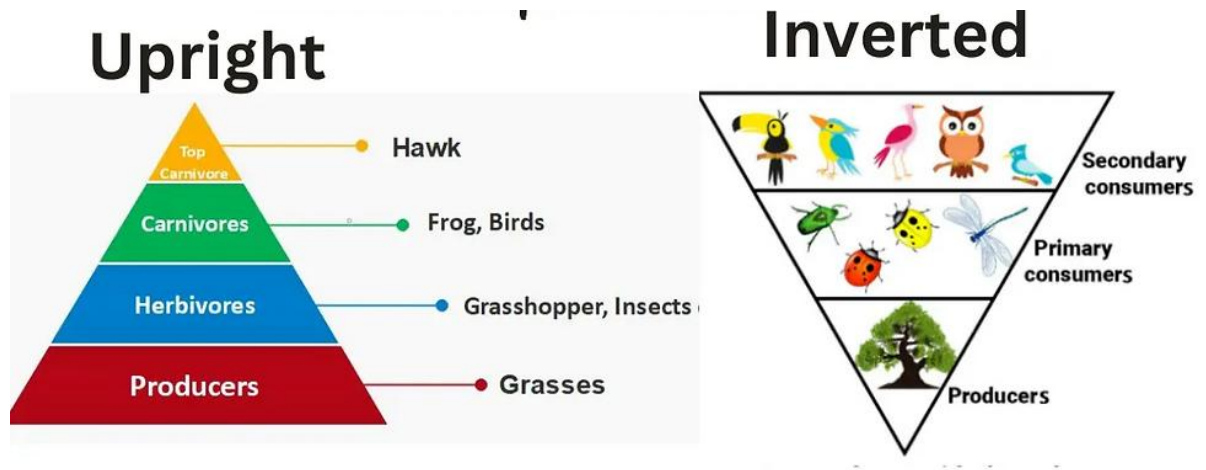
ECOLOGICAL PYRAMIDS

In an ecosystem, green plants – the **producers**, utilize energy directly from sunlight and convert it into matter. A large number of these organisms form the most **basic, or first ‘trophic level’** of the food pyramid. The **herbivorous** animals that eat plants are at the **second trophic level** and are called primary consumers. The **predators** that feed on them form the **third trophic level** and are known as secondary consumers. Only a few animals form the third trophic level consisting of **carnivores** at the **apex** of the food pyramid. This is how energy is used by living creatures and flows through the ecosystem from its base to the apex. Much of the energy is used up in activities of each living organism.



Forms of Ecological Pyramid

1. **Upright:** In terrestrial ecosystems, where producers (green plants) have more biomass than consumers.
2. **Inverted:** In aquatic ecosystems, where the mass of producers is lower than the mass of heterotrophs like fish and insects. In a parasite's food chain, the pyramids are inverted.



Types of Ecological pyramid

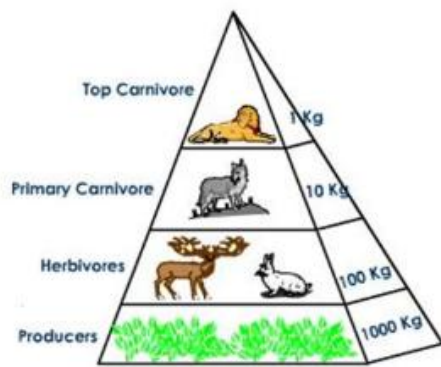
Ecological pyramids are of three general types as under:

1. Pyramid of numbers

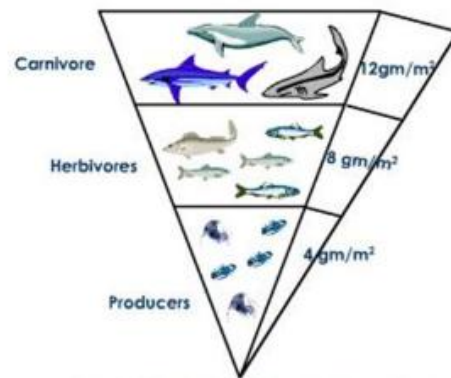
The pyramids of numbers show the relationship between producers, herbivores and carnivores at successive trophic levels in terms of their numbers. For an upright pyramid, **grassland** the producers, which are mainly grasses, are always **maximum** in number. The number shows a decrease towards apex. In a **parasite's food chain**, the pyramids are inverted. This is for the reason that a single plant may support the growth of many herbivores. In its turn, each herbivore may provide nutrition to several parasites, which support many hyperparasites. Consequently, from the producer towards consumers, there is a reverse position. In other words, the number of organisms gradually shows an increase, making the pyramid inverted in shape.

2. Pyramids of biomass

The pyramids of biomass in different types of ecosystems may be compared as in case of **grassland** and forest there is generally a gradual decrease in biomass of organisms at successive levels from the producers to the top carnivores. In this way, the pyramids are upright. However, in a **pond** the producers are small organisms, their biomass is least, and this value gradually shows a decrease towards the apex of the pyramid and the pyramids are made inverted in shape.



Upright Pyramid of biomass in a Terrestrial Ecosystem



Inverted Pyramid in an Aquatic Ecosystem

3. Pyramid of energy

The energy pyramid gives the best picture of overall nature of the ecosystem. Here, number and weight of organisms at any level depends on the rate at which food is being produced. Comparing the pyramid of energy with the pyramids of numbers and biomass, which are pictures of the standing situations (organisms present at any moment), the pyramid of energy is a picture of the rates of passage of food mass through the food chain. It is always upright in shape

BIODIVERSITY

The word biodiversity is a combination of two words: “biological and diversity” and refers to the variety of life on the Earth. Biodiversity means the variety of all living things on Earth, including plants, animals, and microorganisms, as well as the ecosystems they form. It’s important because it helps nature stay balanced and supports life by providing food, clean water, and air. Even though it is impossible to access the number of species of living things on earth an approximate estimate puts it at 10 to 80 million species.

Biodiversity is generally described in terms of its 3 fundamental and hierarchically related levels of biological organisms. They are:

1. Genetic diversity

It includes the genetic variations within species, both among geographically separated populations and among individuals within single population. This is the difference in the genetic makeup in one individual species. i.e. variability within the species. For example, in dogs, there are many breeds like Labradors, Poodles, and Bulldogs, all of which belong to the same species but have different genetic traits such as size, fur type, and behavior.

2. Species diversity

This is the variety or variance of distinct types of living organisms in different habitats. It includes full range of species from micro-organisms to giants and mammoth varieties of plants and animals. Example - single celled viruses and bacteria etc. Multi-cellular plants, animals and fungi.

3. Ecosystem diversity

It studies variation in the biological communities in which species live, exist and interact. This includes the different types of forests, grass lands, wet lands, water bodies like streams, lakes and oceans, coral reefs, rocky-mountains etc.

4. Functional Diversity

This includes the different types of biological and chemical processes such as mass and energy flow essential for the survival of living organisms.

Importance of Biodiversity

Biological diversity is very important for the existence of the human race. In addition to the practical importance, Biodiversity lends aesthetic beauty to nature. The benefits arising from biological diversity are:

- Protection of water resources like maintenance of hydrological cycles, regulating and stabilising water runoff, and buffer against extreme events such as flood and drought.
- Soils formation and protection: Formation and maintenance of soil structure and the retention of moisture and nutrient levels.
- Nutrient storage and cycling: Plants take up nutrients from the soil as well as from the air, and these nutrients can then form the basis of food chains, to be used by a wide range of other life forms. Pollution breakdown and absorption
- Breakdown and absorption of pollutants: Ecosystems and components of ecosystems from bacteria to higher life forms are involved in breakdown and assimilative processes of pollutants. Natural and artificial wetlands are being used to filter effluents to remove nutrients, heavy metals and suspended solids, reduce the biochemical oxygen demand and destroy potentially harmful microorganisms.
- Contribution to climate stability: Vegetation influences climate at the macro and micro levels. Undisturbed forest helps to maintain the rainfall in its immediate vicinity by recycling water vapour at a steady rate back into the atmosphere.
- Recovery from unpredictable events: Healthy ecosystems improve the chances of recovery of plant and animal populations from unpredictable natural catastrophic events such as fire, flood and cyclones and from disasters caused by humans. Biological Resources
- Food : Existence of human beings and that of most other organisms is heavily dependent on primary producers, mainly plants.
- Medicinal resources: Plant and animal products have long since been used as medicines due to their curative properties. Indian and Chinese medical systems make elaborate use of herbs and herbal products.
- Wood products: Wood is a basic commodity used worldwide. It is a primary source of fuel, is used in construction, and forms the basis for paper production.

- Ornamental plants: Ornamental and horticultural purposes, for providing aesthetic beauty
- Breeding stocks, population reservoirs: Natural areas provide support systems for commercially valuable resources.
- Future resources: The wild and unknown population of plants and animals offer enormous scope for research and development in areas of agriculture, industry and medicine, social benefits
- Research, education and monitoring: Natural areas provide excellent living laboratories for research in ecology and evolution.
- Recreation: Diverse biological assemblies like animal habitats, forests, mountains, parks and gardens, coastal areas and beaches, provide aesthetic beauty to our environment.
- Cultural values: The aesthetic values of our natural ecosystems and landscapes contribute to the emotional and spiritual well-being of human beings.

Threats to biodiversity

Threats to biodiversity are factors that harm the variety of life on Earth. Some key threats impact biodiversity:

1.Habitat Loss: The primary cause of loss of biodiversity is habitat destruction, resulting from expansion of human population. When forests are cut down, wetlands drained, or oceans polluted, the natural homes of plants and animals are destroyed. This forces species to move, adapt, or face extinction. Fragmentation of landscape by construction of roads, infrastructure also results in biodiversity loss. Environmental fluctuations, disease outbreak leads to extinction of small isolated habitats. Example - Polar bear losing sea ice or orangutans losing rainforests to palm oil.

2.Invasive Species: Non-native species introduced to an area can outcompete, prey on, or spread diseases to native species. The introduction of exotic or invasive species is a significant reason of extinction. Great majority of invasive species do not become established in the new environment. Alien species become 'invasive' when they displace the native species and upset the ecological balance. Some successful exotic species may kill or eat native species to the point of extinction; or may alter the habitat that many natives are no longer able to persist. Non – native invasive species are particularly destructive on islands. For example, the cane toad in Australia has harmed many local species by disrupting the ecosystem.

3.Overexploitation: Overfishing, hunting, or harvesting plants beyond sustainable levels reduces populations, sometimes driving species to extinction, like the dodo bird due to excessive hunting.

MAN AND ENVIRONMENT

The physical environment directly or indirectly affects our behaviour. For example, it has been observed that people become more irritable and aggressive when the weather is hot

and humid, in comparison to cold weather. You must have read in the daily newspapers about the increase in “road rage” cases during the summer months. Our interest in such environmental variables has led to the development of a field known as Environmental Psychology. This field of psychology is dedicated to the study of reciprocal relationships between psychological processes and physical environments, both natural and man made. Reciprocal relationship deals with two-way process in which environment influences human behaviours and human beings affect environment. In order to understand various aspects of this interaction it would be useful to understand various types of environment which we encounter.

A brief description of major types of environment is given below:

1. Physical environment : It includes both physical reality and social-cultural phenomenon that surround us. The noise, the temperature, the quality of air and water, and various objects and things constitute the physical world around us. The social and cultural environment: It includes the aspects of social interaction including its products such as beliefs, attitudes, stereotypes, etc. The material and non-material aspects of environment are included in it.

2. Psychological Environment: It includes the perceptions and experiences pertaining to any environmental setting. Some environments may be stimulating and exciting while others may be dull and boring. Expression of psychological is often used in the organizational context.

The human-environment interaction has five major components. These components are briefly described below:

1. Physical Environment: It includes aspect of natural environment such as climate, terrain, temperature, rainfall, flora, fauna, etc.
2. Social –Cultural Environment: It includes all aspects of cultural environment such as norms, customs, process of socialization, etc. It includes all the aspects dealing with other people and their creations.
3. Environmental Orientations: It refers to the beliefs that people hold about their environment. For example, some people hold environment equivalent to God and therefore they perceive all its aspects with respect and reverence and try to maintain it in a perfect form and do not degrade it.
4. Environmental Behaviour: It refers to the use of environment by people in the course of social interactions. For example, considering the environment as personal space, where the individual identifies himself with it.
5. Products of Behaviour: These include the outcomes of people’s actions such as homes, cities, dams, schools, etc. That is, these are products or outcomes dealing with the environment.

HEALTH AND ENVIRONMENT

Environmental health refers to components of human health and the quality of the environment. “It also refers to the theory and practice of assessing, correcting, controlling, and preventing those factors in the environment that can potentially affect the health of the present and future generations” (WHO, 1993). Environmental health aims to control disease and illness. It also aims to promote the well-being of humans by analysing the sources and hazardous agents in the environment. Environmental health services is defined as “those services which implement environmental health policies through monitoring and

control activities” (WHO, 1993). This can be achieved by the improving our surroundings, maintain cleanliness and follow ecofriendly technologies. The professionals working in this discipline are known as environmental health officers or specialists, public health inspectors, and practitioners. Researchers and policy-makers also are needed in the field of environmental health. Health is now not only the responsibility of medical personnels, nursing and paramedical staff. It is every individual’s responsibility, the government, and NGO’s to work together and bring about the required change. Such changes can bring about good quality of the environment and affect the human health.

Environment and health are not separate entities. Both are driven by external forces such as population and poverty. For a healthy nation the government should focus on the health of its people and maintain a positive balance between human activities and the environment. For achieving these goals the following are important.

They include:

- (a) Economic development must meet people’s needs
- (b) Ecological sustainability must be achieved.

Therefore joint action is needed at the local, national and global level by individuals and between governmental and nongovernmental agencies. We can create healthy environments in a number of ways. This can be done by greening the environment and by managing wastes efficiently. Human activities like burning or emissions from industries can affect health, as toxic chemicals are released into the atmosphere polluting our physical environment.

ENVIRONMENTAL ETHICS

Environmental ethics is the branch of ethics that deals with the relationship between humans and the natural world. It asks questions like:

- How should we treat the environment?
- What responsibilities do we have toward nature, animals, and future generations?
- Is it right to exploit nature for our own benefit, or should we protect it?

Environmental ethics is a new sub-discipline of philosophy that deals with the ethical problems surrounding environmental protection. It aims to provide ethical justification and moral motivation for the cause of global environmental protection. There are several distinctive features of environmental ethics that deserve our attention.

SUSTAINABLE DEVELOPMENT

Sustainable development is defined as development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It includes social development and economic opportunity on one hand and the requirements of environment on the other. It is based on improving the quality of life for all, especially the poor and deprived within the carrying capacity of the supporting ecosystems. The concept of Sustainable Development is illustrated in the **Venn diagram**.



Pillars of Sustainable Development

1.Social Sustainability

Social sustainability includes environmental justice, human health, resource security, and education, among other important social elements of society. Under the three pillars concept, efforts to promote social sustainability should also aim to foster economic and environmental benefits.

For businesses, efforts to generate social sustainability could include focusing company efforts on employee retention instead of economic priorities. For example, investments in the well-being of employees are likely to generate economic benefits for the company by increasing employee motivation.

Efforts to increase social sustainability can also benefit the environment. For example, people's diet choices can have a substantial impact on both human health and the health of the environment, therefore advocacy for healthier eating can benefit the environment, too.

2.Economic Sustainability

Economic sustainability includes job creation, profitability, and proper accounting of ecosystem services for optimal cost-benefit analyses. When it comes to the job market, research shows high rates of employment benefit both the economy and the people's social well-being through the resource security employment provides. In this way, the economic drivers requiring companies to need employees and people to need jobs can also foster social sustainability if employment offers people security.

However, today's gig economy places social and economic sustainability at odds with one another. The gig economy causes many people to contribute to the economic sustainability of companies without receiving the social safety nets typically provided by employment in return.

Efforts to be more environmentally sustainable can also benefit the economic sustainability of an organization. For example, recycling valuable materials, such as electronic waste and textile waste, can lower operating costs and reduce the intensity of resource extraction required to sustain businesses.

3.Environmental Sustainability

Environmental sustainability focuses on the well-being of the environment. This pillar includes water quality, air quality, and reduction of environmental stressors, such as greenhouse gas emissions. Human health depends greatly on the quality of a person's environment, inextricably linking human health and the state of the environment. Therefore, efforts to preserve and restore the environment benefit people, too.

The environment also provides natural resources necessary to foster economic sustainability. Companies rely on the extraction of natural resources to be economically sustainable. Efforts to extract resources at levels that are sustainable for the environment will also provide economic sustainability through the continued availability of resource.

Need for Sustainable development

- Sustainable development is crucial to balance economic growth, environmental protection, and social equity, ensuring that resources are available for future generations
- It helps address critical issues like climate change, biodiversity loss, and pollution while fostering long-term economic stability
- By promoting renewable energy, efficient resource use, and eco-friendly practices, it ensures a healthier planet
- Sustainable development also uplifts marginalized communities, creating inclusive opportunities and reducing inequalities
- It safeguards the well-being of both present and future populations by prioritizing harmony between human activity and the natural world

Sustainable Development Goals (SDGs)

The SDGs were formally adopted at the **UN Sustainable Development Summit in September 2015** as part of the **2030 Agenda for Sustainable Development**. The Sustainable Development Goals (SDGs) are a set of 17 global goals established by the United Nations in 2015 as part of the 2030 Agenda for Sustainable Development. They aim to address urgent global challenges, promote prosperity, and protect the planet while ensuring no one is left behind



SUSTAINABLE DEVELOPMENT GOALS

1 NO POVERTY



2 ZERO HUNGER



3 GOOD HEALTH AND WELL-BEING



4 QUALITY EDUCATION



5 GENDER EQUALITY



6 CLEAN WATER AND SANITATION



7 AFFORDABLE AND CLEAN ENERGY



8 DECENT WORK AND ECONOMIC GROWTH



9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



10 REDUCED INEQUALITIES



11 SUSTAINABLE CITIES AND COMMUNITIES



12 RESPONSIBLE CONSUMPTION AND PRODUCTION



13 CLIMATE ACTION



14 LIFE BELOW WATER



15 LIFE ON LAND



16 PEACE, JUSTICE AND STRONG INSTITUTIONS



17 PARTNERSHIPS FOR THE GOALS

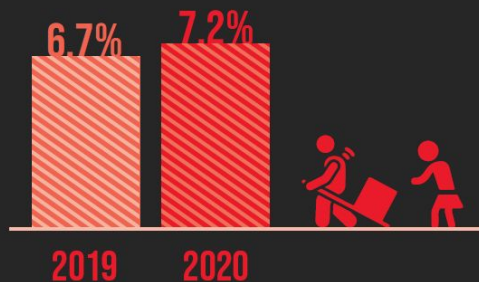


SUSTAINABLE
DEVELOPMENT
GOALS

1 NO POVERTY



WORKING POVERTY RATE ROSE FOR THE FIRST TIME IN TWO DECADES



PUSHING AN ADDITIONAL
8 MILLION WORKERS
INTO POVERTY

WHY IS THERE SO MUCH POVERTY IN THE WORLD?

POVERTY HAS MANY DIMENSIONS, BUT ITS CAUSES INCLUDE UNEMPLOYMENT, SOCIAL EXCLUSION, AND HIGH VULNERABILITY OF CERTAIN POPULATIONS TO DISASTERS, DISEASES AND OTHER PHENOMENA WHICH PREVENT THEM FROM BEING PRODUCTIVE.

SOME TARGETS TO ACHIEVE BY 2030

- ERADICATE EXTREME POVERTY FOR ALL PEOPLE EVERYWHERE, CURRENTLY MEASURED AS PEOPLE LIVING ON LESS THAN \$1.25 A DAY.
- ENSURE THAT ALL MEN AND WOMEN, IN PARTICULAR THE POOR AND THE VULNERABLE, HAVE EQUAL RIGHTS TO ECONOMIC RESOURCES, AS WELL AS ACCESS TO BASIC SERVICES.
- ENSURE SIGNIFICANT MOBILIZATION OF RESOURCES FROM A VARIETY OF SOURCES, IN ORDER TO PROVIDE ADEQUATE AND PREDICTABLE MEANS FOR DEVELOPING COUNTRIES, IN PARTICULAR LEAST DEVELOPED COUNTRIES, TO IMPLEMENT PROGRAMMES AND POLICIES TO END POVERTY EVERYWHERE.

2 ZERO HUNGER



CONFLICT, COVID-19, CLIMATE CHANGE
AND GROWING INEQUALITIES
ARE CONVERGING TO UNDERMINE
FOOD SECURITY WORLDWIDE



WHY ARE THERE SO MANY HUNGRY PEOPLE?

ALONG WITH CONFLICT, CLIMATE SHOCKS AND THE LOCUST CRISIS, THE PANDEMIC POSES AN ADDITIONAL THREAT TO FOOD SYSTEMS. CIVIL INSECURITY AND DECLINING FOOD PRODUCTION HAVE ALL CONTRIBUTED TO FOOD SCARCITY AND HIGH FOOD PRICES.

SOME TARGETS TO ACHIEVE BY 2030

- END HUNGER AND ENSURE ACCESS BY ALL PEOPLE, IN PARTICULAR THE POOR AND PEOPLE IN VULNERABLE SITUATIONS, INCLUDING INFANTS, TO SAFE, NUTRITIOUS AND SUFFICIENT FOOD ALL YEAR ROUND.
- ENSURE SUSTAINABLE FOOD PRODUCTION SYSTEMS AND IMPLEMENT RESILIENT AGRICULTURAL PRACTICES THAT INCREASE PRODUCTIVITY AND PRODUCTION THAT HELP MAINTAIN ECOSYSTEMS AND IMPROVE THEM.
- ADOPT MEASURES TO ENSURE THE PROPER FUNCTIONING OF FOOD COMMODITY MARKETS AND THEIR DERIVATIVES AND FACILITATE TIMELY ACCESS TO MARKET INFORMATION, INCLUDING ON FOOD RESERVES, IN ORDER TO HELP LIMIT EXTREME FOOD PRICE VOLATILITY.



#3

Good Health And Well Being

- ☐ Ensuring healthy lives and promoting well being at all ages
- ☐ Eradicate a wide range of diseases and address many different persistent and emerging health issues
- ☐ Increase the life expectancy and decrease child and maternal mortality
- ☐ Need efficient funding in health systems
- ☐ Improved sanitation and hygiene
- ☐ Increased access to physicians
- ☐ Prevention and treatment of substance abuse(narcotics and alcohol)
- ☐ Research and development of vaccines and medicines for the various communicable and non-communicable diseases



#4

Quality education

- ❑ Ensuring inclusive and equitable quality education and promote lifelong learning opportunities for all
- ❑ All girls and boys complete free primary and secondary schooling by 2030
- ❑ Equal access to affordable vocational training
- ❑ Eliminate gender and wealth disparities and achieve universal access to a quality higher education
- ❑ Good teaching and learning processes
- ❑ Effective environment for learning with extra-curricular activities



#5

Gender Equality

5 GENDER
EQUALITY



- ❑ “Achieve gender equality and empower all women and girls”
- ❑ End all forms of discrimination against women and girls everywhere.
- ❑ Eliminate all forms of violence against all women and girls and other types of exploitations.
- ❑ Eliminate all harmful practices, such as child, early and forced marriage etc.
- ❑ Recognize and value unpaid care and domestic work.
- ❑ Ensure women’s full and effective participation in political, economic, and public life.
- ❑ Ensure access to reproductive health rights as agreed in the Programme of Action of the ICPD
- ❑ Give women equal rights to resources, ownership of property, financial services, inheritance etc

#6

Clean Water And Sanitation

6 CLEAN WATER
AND SANITATION



- ☐ “Clean area and sanitation for all”
- ☐ Safe and affordable drinking water.
- ☐ End open defecation and provide access to sanitation and hygiene.
- ☐ Improve water quality, wastewater treatment, and safe reuse.
- ☐ Increase water-use efficiency and ensure freshwater supplies.
- ☐ Implement integrated water resource management.
- ☐ Protect water-related ecosystems.
- ☐ Expand water and sanitation support to developing countries.
- ☐ Support local engagement in water and sanitation management.

#7

Affordable and clean energy

- It aims to "Ensure access to affordable, reliable, sustainable and modern energy for all."
- Energy remains one of the critical challenges of the future.
- The goal has five targets to be achieved by 2030:
 - 1.Universal access to modern energy.
 - 2.Increase global percentage of renewable energy.
 - 3.Double the improvement in energy efficiency.
 - 4.To promote access to research,technology and investments in clean energy.
 - 5 Expand and upgrade energy services for developing countries.

7 AFFORDABLE AND
CLEAN ENERGY





#8

Decent work and economic growth

- The full title is to "Foster sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all".
- The COVID-19 pandemic is pushing the world into the worst global economic crisis .
- This goal aims at ensuring the economic sector of every country provides the necessary need for its citizen to have a good life irrespective of their background, race or culture.

#9 Industry, Innovation and Infrastructure

9 INDUSTRY, INNOVATION
AND INFRASTRUCTURE



- Global Manufacturing has rebounded from the pandemic
But LDCs are left behind
- Higher Technology Industries are far more resilient in crises
- Small scale industries still lack necessary financial aid
- Passenger

#10 Reduce inequality within and among countries

- Pandemic Caused first income disparity
- Many migrants lost their life
- This has led to discrimination
- Global refugee count has increased
- War in ukraine has pushed the above factor



#11

Sustainable Cities and Communities

11 SUSTAINABLE CITIES
AND COMMUNITIES



- Make cities and human settlement inclusive, safe, resilient and sustainable
- Cities are already responsible for 70 percent of global waste and consume almost 80 percent of world's energy.
- How to make them sustainable?
 - * Promoting urban agriculture.
 - * Encouraging healthy diets.
 - * Reducing and managing food waste.
 - * Boosting green spaces for healthier environment and lifestyle.
 - * Reconnecting cities with surrounding rural areas.

#12

Responsible Consumption and Production.



- Our planet has provided us with an abundance of natural resources but we have not utilised them responsibly.
- Sustainable consumption and production is about :
 - * Promoting resource and energy efficiency.
 - * Providing access to basic services.
- How to ensure this?
 - * Sustainable management and use of natural resources.
 - * Reduce waste(Reduce,Reuse,Recycle).
 - * Conscious consumption.
- Unsustainable patterns of consumption leads to climate change,biodiversity loss, pollution.



#13

Climate action

- ☐ Take urgent action to combat climate change and its impacts
- ☐ Urgent action to address both the pandemic and the climate emergency
- ☐ Strengthen the abilities of countries to cope and deal with the impacts of climate change
- ☐ Limit global temperature
- ☐ Turn to renewable energy and increase adaptation efforts
- ☐ Must move towards a low carbon economy
- ☐ Integrate climate change measures into policies and planning

#14

Life below water

14 LIFE
BELOW WATER



- The official wording is to "Conserve and sustainably use the oceans, seas and marine resources for sustainable development".
- Oceans and fisheries support the global population's economic, social and environmental needs.
- Over 3 billion people depend on marine life for their livelihood.
- However there has been a 26 percent increase in acidification since the industrial revolution.
- Effective strategies are needed to advance the sustainable use of oceans.

#15

LIFE ON LAND



- Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.
- Ensure the conservation, restoration and sustainable use of terrestrial and inland freshwater ecosystems and their services.
- Combat desertification, restore degraded land and soil.
- Ensure the conservation of mountain ecosystems, including their biodiversity, in order to enhance their capacity to provide benefits that are essential for sustainable development.
- Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity.
- Take urgent action to end poaching and trafficking of protected species of flora and fauna and address both demand and supply of illegal wildlife products.
- Integrate ecosystem and biodiversity values into national and local planning, development processes, poverty reduction strategies and accounts.

#16

PEACE, JUSTICE AND STRONG INSTITUTIONS

16 PEACE, JUSTICE
AND STRONG
INSTITUTIONS



- Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.
- Significantly reduce all forms of violence and related death rates everywhere.
- End abuse, exploitation, trafficking and all forms of violence against and torture of children.
- Promote the rule of law at the national and international levels and ensure equal access to justice for all.
- Substantially reduce corruption and bribery in all their forms.
- Develop effective, accountable and transparent institutions at all levels.
- Ensure responsive, inclusive, participatory and representative decision-making at all levels.
- Strengthen relevant national institutions, including through international cooperation, for building capacity at all levels, in particular in developing countries, to prevent violence and combat terrorism and crime.
- Promote and enforce non-discriminatory laws and policies for sustainable development.
- Provide legal identity for all, including birth registration.
- Broaden and strengthen the participation of developing countries in the institutions of global governance.

#17

PARTNERSHIPS FOR THE GOALS



- Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development.
- Strengthen domestic resource mobilization.
- Mobilize additional financial resources for developing countries from multiple sources.
- Adopt and implement investment promotion regimes for least developed countries.
- Promote the development, transfer, dissemination and diffusion of environmentally sound technologies to developing countries.
- Promote a universal, rules-based, open, non-discriminatory and equitable multilateral trading system under the World Trade Organization.
- Enhance global macroeconomic stability, including through policy coordination and policy coherence.
- Enhance policy coherence for sustainable development.
- Build on existing initiatives to develop measurements of progress on sustainable development that complement gross domestic product, and support statistical capacity-building in developing countries.